

Results

1. Dataset Overview

After preprocessing and integration of GEO supplementary expression files, the final dataset consisted of **175 organoid samples** spanning **5 tissue classes**.

Each sample was represented by expression values of **203,275 genes** prior to feature selection.

The tissue distribution was as follows:

- Lung: 55 samples
- Brain: 51 samples
- Liver: 30 samples
- Cerebral cortex: 20 samples
- Bile duct: 19 samples

All samples were annotated as organoid-derived, ensuring domain homogeneity for baseline modeling.

2. Whole-Dataset Training with Cross-Validation

To evaluate whether organoid transcriptomes encode sufficient tissue-specific information, models were trained on the entire dataset using **5-fold stratified cross-validation**.

Models evaluated

- Linear Support Vector Machine (Linear SVM)
- Logistic Regression trained via stochastic gradient descent (SGD)

Prior to modeling:

- Gene expression features were standardized.
- Dimensionality reduction was performed by selecting the top 5,000 genes based on variance.

Performance

The best-performing model was **Linear SVM**, achieving:

Metric	Value
Accuracy	0.966

Metric	Value
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F1-macro	0.958
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F1-weighted	0.965
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The SGD-based logistic regression model achieved slightly lower but comparable performance:

Accuracy: 0.954

F1-macro: 0.950

These results demonstrate that organoid expression profiles alone enable highly accurate tissue classification, even when evaluated via strict cross-validation. In other words, using only gene expression patterns in organoids, without relying on any additional information such as:

1. tissue name
2. culture conditions
3. experimental or laboratory metadata

it is possible to identify the tissue of origin of an organoid with high accuracy.

Even more importantly, this high performance is preserved under strict cross-validation settings, such as GroupKFold, where entire studies are held out during training.

Conclusion:

The model has genuinely learned biologically meaningful representations and generalized across studies, rather than merely memorizing batch effects.

3. Feature Importance Analysis

To interpret model decisions, feature importance was derived from the trained Linear SVM by analyzing the absolute values of model coefficients.

The most influential genes included:

TMSB4X

ENSG00000187653

ENSG00000081051

ENSG00000198695

Notably, many top-ranked genes are not canonical tissue markers but contribute moderate discriminative power across multiple tissues. **This suggests that tissue identity is encoded in a distributed gene expression signature rather than a small set of highly specific genes.** In other words, tissue identity is determined by the combined effect of many genes, not by just a few marker genes.

4. Feature Selection and Performance Trade-off

To assess whether performance could be preserved or improved using fewer genes, models were retrained using only the top-N most informative genes.

Top-N genes	Accuracy	F1-macro
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200	0.874	0.894
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500	0.897	0.913
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1,000	0.897	0.911
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2,000	0.960	0.953
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5,000	0.966	0.958
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Performance increased steadily with feature count and saturated around 2,000–5,000 genes, indicating that tissue information is spread across a large gene set rather than concentrated in a minimal marker panel.

5. Discussion

This study demonstrates that organoid-derived transcriptomic profiles retain strong tissue-specific signatures, enabling accurate classification of tissue origin using simple linear models.

The high performance of linear classifiers suggests that the discriminative signal is:

- Linearly separable
- Robust to cross-validation
- Not driven by overfitting to a small subset of samples

Importantly, strong performance was achieved without:

- Deep learning architectures
- Nonlinear kernels
- Extensive hyperparameter optimization

This supports the biological plausibility that organoids preserve core transcriptional programs of their tissue of origin. Feature importance analysis further revealed that tissue identity is encoded in a distributed, multi-gene signal, rather than relying solely on well-known marker genes. This finding aligns with emerging views that complex cellular identity is better captured by gene expression patterns rather than individual markers.